How-To Library

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Data Domain Profiling and Data Masking for Hadoop

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Abstract

In TDM, you can use Hadoop connections when you want to lower the cost of raw data storage and solve large scale analytics by using the distributed computing feature of Hadoop. You can perform data domain profiling, data masking, and data movement on Big Data Edition Hadoop clusters.

This article describes how to perform data domain profiling and data masking on a relational database and then move the data in to the target Hive database.

Supported Versions

- Test Data Management 9.7.0

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Overview

You can perform data masking and data domain profiling for Hadoop.

Use Hadoop sources to lower the cost of raw data storage and to solve large scale analytics by using the distributed computing feature of Hadoop. For example, when you move sensitive data into Hadoop, you can classify data for analytics, data provisioning for testing, or any other purposes.

The analytics results processed on Hadoop are faster and cost-effective, and you can extract the analytics results to a conventional database.

Scenario

You work for an organization in the software testing department, and you want to ensure that raw sensitive data is moved into Hadoop and is made available to testers to build analytical reports on Hive. You need to profile the data to identify sensitive columns, such as credit card numbers, and mask them. The resulting test data must have data integrity.

To perform data domain profiling and data masking operations for Hadoop, you must perform the following tasks:

1. Create a Hive database target connection.
2. Create a credit card masking rule and add to a project.
3. Create a data domain.

4. Create and run a data domain profile to identify sensitive data, such as credit card numbers.
5. Assign the credit card masking rule to the sensitive column.
6. Create a Hadoop plan.
7. Generate and run the plan.

Prerequisites

Before you begin, you must perform the following tasks:

1. Install and configure TDM.
2. Create an Oracle database source connection.
3. Create a project and import data sources.

Step 1. Create a Target Hive Database Connection

In Test Data Manager, you can configure the Hive properties, create a Hive database connection, and use the connection as the target.

1. Log in to Test Data Manager and click **Administrator** > **Preferences**.
2. In the Hive Properties section, click **Edit**.

   The **Edit-Preferences** dialog box appears.
3. Select a Hive connection to run the Hadoop plan.

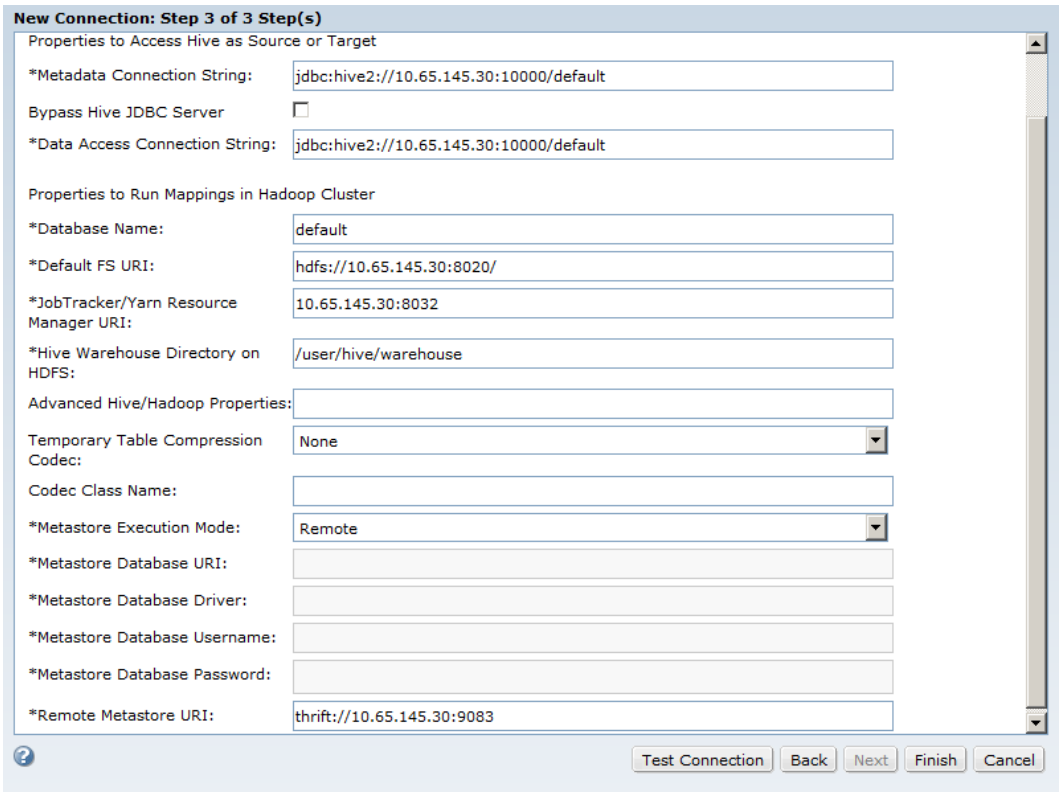
   Note: You cannot use this connection as target connection in the Hadoop plan.
4. To store the mapping in the Model repository for future use, select **Persist Mapping**.
5. Click **OK**.
6. Click the **Connections** tab.
7. Click **Actions** > **New Connection**

   A tab opens to display the new connection properties.
8. Select the **Hive** connection type.
9. Enter the connection name, description, and owner information.

   The connection name must begin with an alphabetic character. If you enter a connection name that begins with a numeric character, a workflow that you use the connection in might fail.
10. Click **Next**.
11. To use Hive as a target, select **Access Hive as a source or target**.
12. To use the connection to run mappings in the Hadoop cluster, select **Use Hive to run mappings in Hadoop cluster**.
13. Enter the user name and environment SQL details.
14. Click **Next**.
15. To access the metadata from the Hadoop cluster, enter the metadata connection string.
16. To access data from the Hadoop cluster, enter the data access connection string.
17. To run mappings in Hadoop cluster, enter the following parameters:
    - Database Name. Enter the name default for tables that do not have a specified database name.

- Default FS URI. Enter the URI to access the default HDFS.
- JobTracker/Yarn Resource Manager URI. Enter the specific node in the Hadoop cluster.
- Hive Warehouse Directory on HDFS. Enter the HDFS file path of the default database.
- Metastore Execution Mode. To connect to a remote metastore, select **Remote**.
- Metastore Database URI. To access metadata in a remote metastore setup, specify the metastore URI with the thrift server details.

The following image shows the Hive connection properties:



18. To test the connection, click **Test Connection**.
19. To save the connection, click **Finish**.

    The connection is visible in the **Administrator | Connections** view.

Step 2. Create a Credit Card Masking Rule and Add to Project

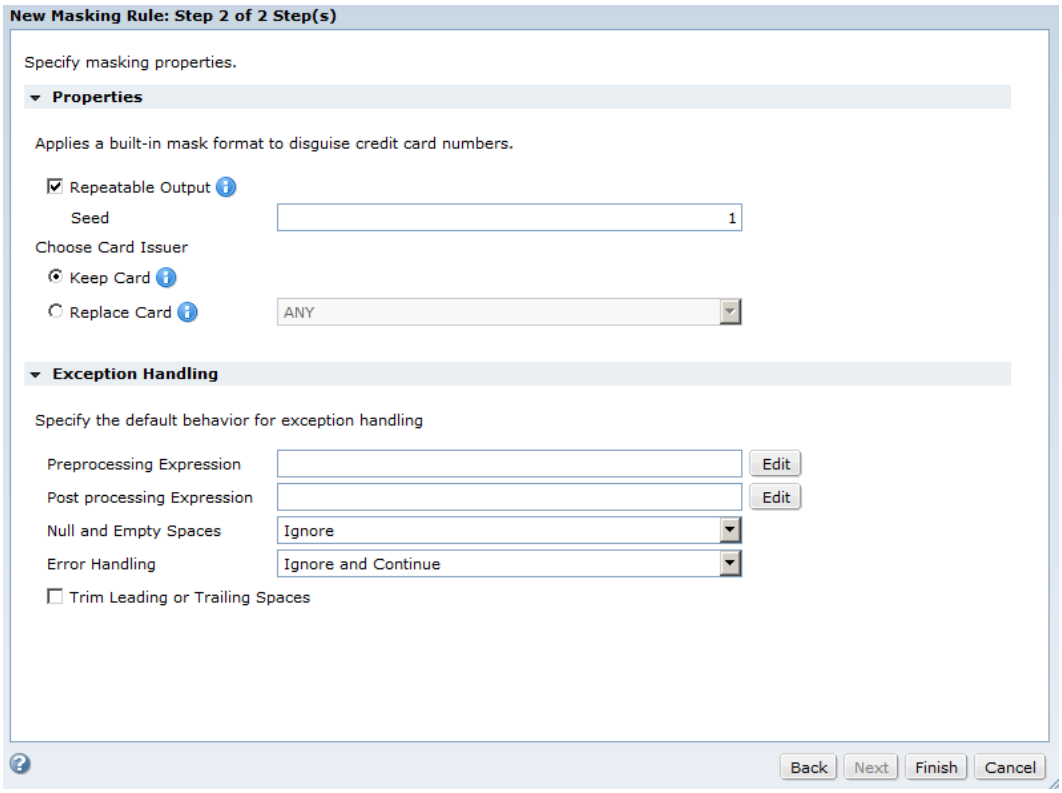
You create a credit card masking rule to mask the credit card numbers. Add the masking to a project.

1. To access the **Policies** view, click **Policies**.
2. Click **Actions** > **New** > **Masking Rule**.

   The **Rule Wizard** appears.
3. Enter a name and optional description for the rule.
4. Select the **String** data type.
5. Choose **Standard** and select the **Credit Card** masking rule from the list.
6. To enable users to override masking parameters for a rule, select the **Override Allowed** option.

7. Click **Next**.
8. Select **Repeatable Output** and enter a seed value.
9. Select **Keep Card** to return the same credit card type for the masked credit card.
10. Enter the exception handling options. Configure how to handle null or empty spaces. Configure whether to continue processing on error.

    The following image shows the credit card masking rule parameters:



11. Click **Finish**.
12. To add the masking rule to the project, click **Projects**.

    You can see a list of projects.
13. Open the project that you created.

    The project window opens in a separate tab.
14. Click **Overview** > **Policies**.
15. Click **Actions** > **Add Additional Rules**.

    The **Add Additional Rules** dialog box appears.
16. Select the credit card masking rule from the list.
17. Click **OK**.

    The masking rule appears under the **Additional Rules** list.

Step 3. Create a Data Domain

When you create a data domain, you can add the masking rules and enable the default masking rule.

1. To create a data domain, click **Policies**.

   The **Policies** view shows a list of the policies, data domains, and rules in the TDM repository.
2. Click **Actions** > **New** > **Data Domain**.
3. Enter a name, sensitivity level, and description for the data domain. Click **Next**.
4. Click **Next**.
5. Optionally, enter a regular expression to filter columns by data pattern.
6. Click **Next**.
7. To skip adding regular expression for metadata, click **Next**.
8. To add preferred masking rules to the data domain, click **Add Rules**.

   The **Add Rules** dialog box appears.
9. Select the credit card data masking rule that you created.
10. Click **OK**.
11. Enable the rule as the default masking rule.
12. Click **Finish**.

Step 4. Create and Run a Data Domain Profile

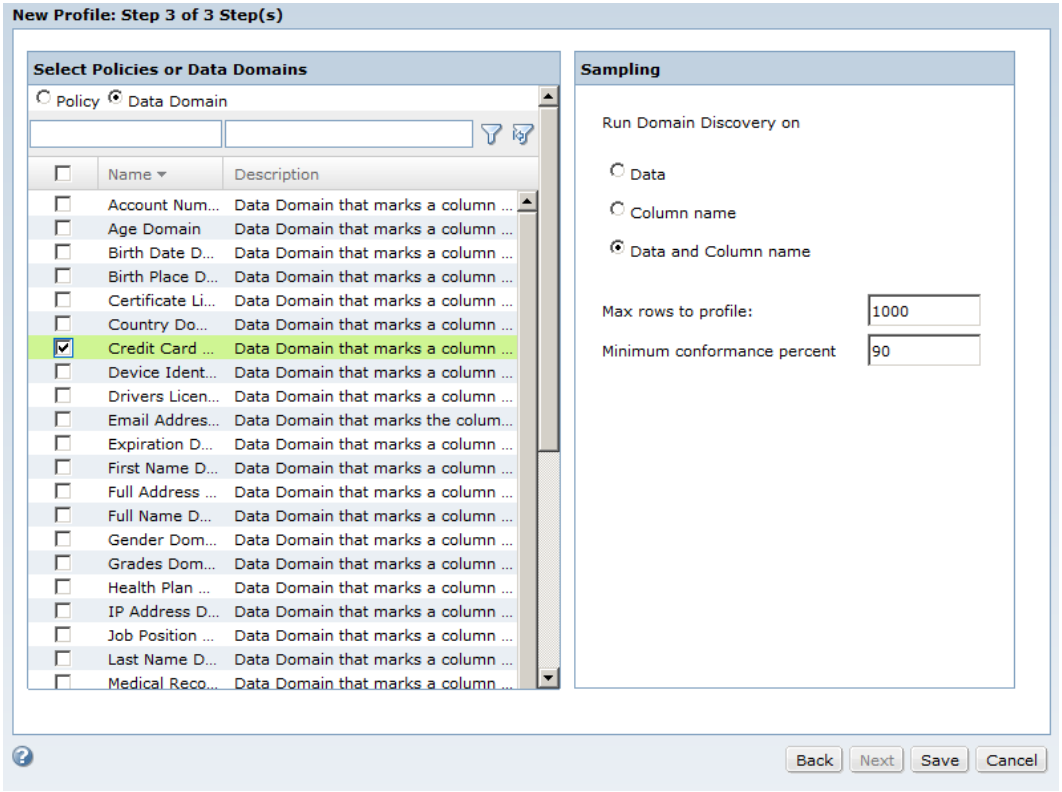
Create and run data domain profiles in the Discover view. A project must contain policies before you create a data domain profile. The policies contain the data domains that you can use in a profile for data discovery.

1. Open the project and click the **Discover** view.
2. Click the **Profile** view.

   The **Profile** view shows a list of the profiles in the project.
3. To create a new profile, click **Actions** > **New Profile**.
4. In the **New Profile** dialog box, enter the profile name and description.
5. Choose to create a data domain profile.
6. Click **Add Tables** and select the tables to profile. Click **OK**.
7. Click **Next.**
8. Select **Data Domain**.
9. Select the data domain that you created.
10. In the **Sampling** panel, select Data and Column name to run data discovery.
11. Enter the maximum number of rows to profile.
12. Enter the minimum conformance percent.

    All rows might not conform to the data domain expression pattern. You can enter a minimum percentage of the profiled rows that must conform.

The following image shows the list of data domains and sampling option that you can configure:



13. Click **Save.**

    The profile opens in a tab.

14. To run the profile, click **Actions** > **Execute**.

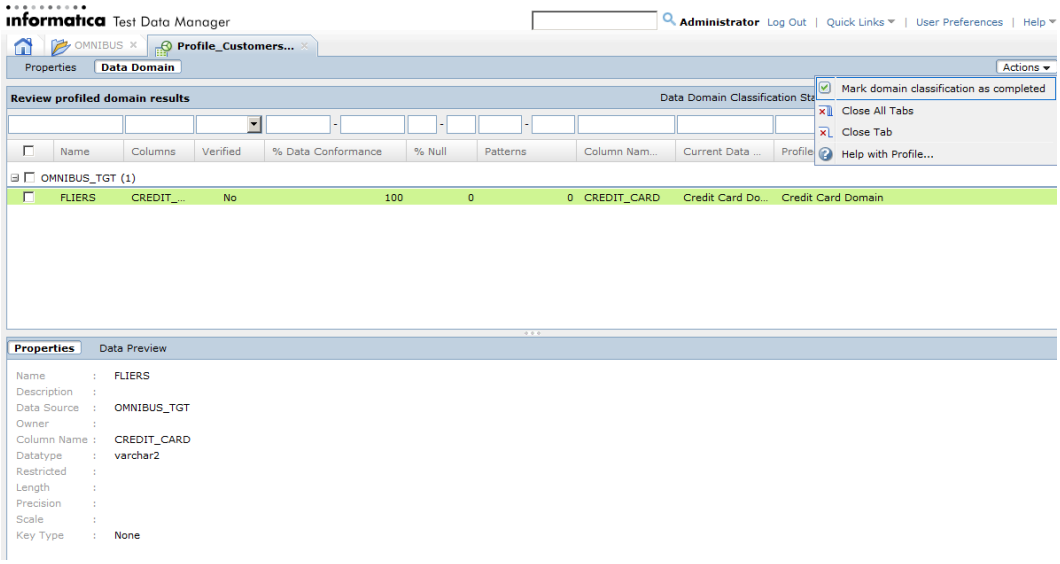
15. In the **Execute Profile** dialog box, click **Execute**.

    View the progress of the profile run from the **Monitor** tab within the project.

16. Open the profile and click the **Data Domain** tab.

17. Click **Actions** > **Mark domain classification as completed**.

    The following image shows the data domain profile review:



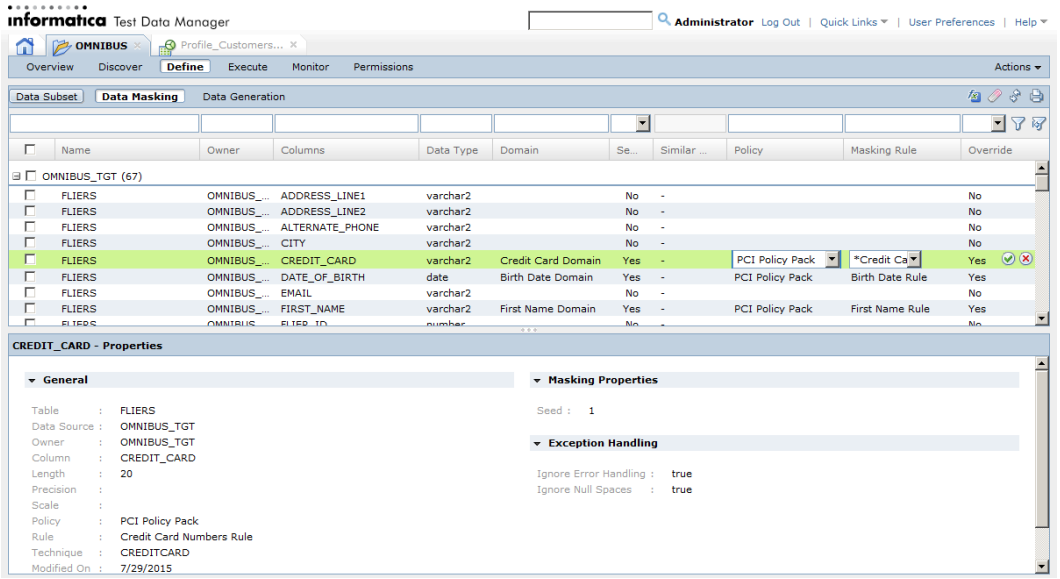
Step 5. Assign the Credit Card Masking Rule

Assign the masking rule from the data domain to the sensitive column in the project source that you want to mask.

1. In the project, click **Define** | **Data Masking** to access the **Data Masking** view.
2. Select the CREDIT_CARD column to assign the masking rule to.
3. Click inside the **Masking Rule** column to view the list of available rules.

   The data domain preferred rules appear at the top of the list.
4. Select the credit card masking rule that you created.

   The following image shows the data masking rule assignment to the CREDIT_CARD column:



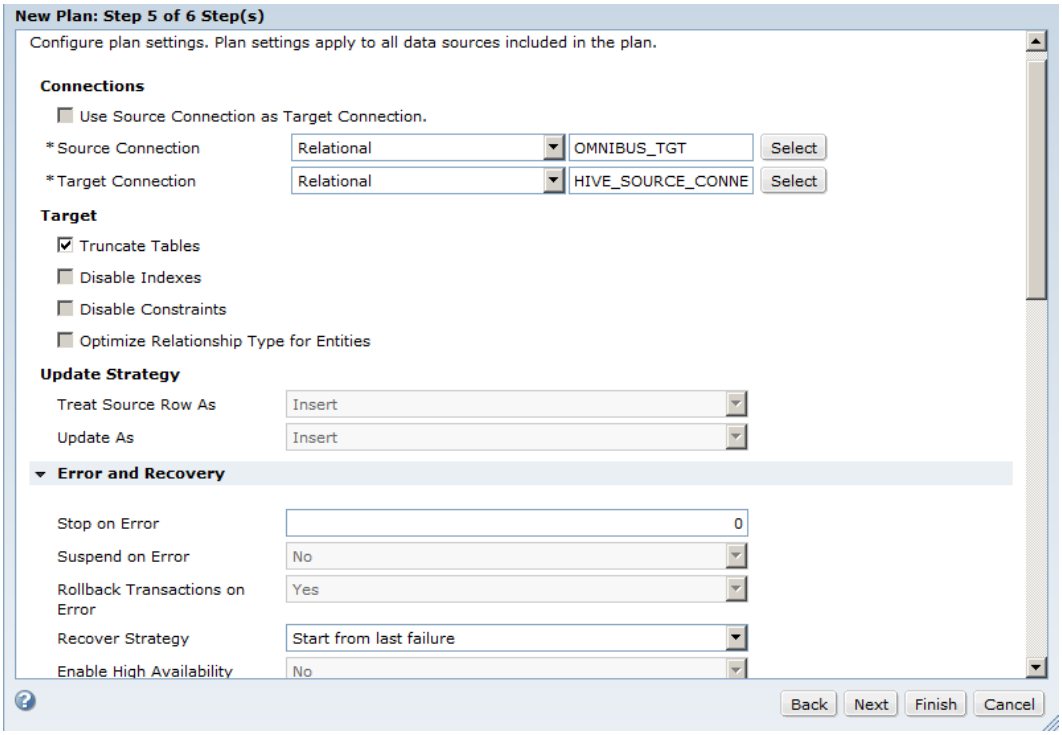
5. To save the assignment, click **Save**.

Step 6. Create a Hadoop Plan

To perform data domain profiling and data masking operations for Hadoop connections, you can create a Hadoop plan. Add data masking components to the Hadoop plan.

1. Open a project and click **Execute**.
2. Click **Actions** > **New**.
3. In the **New Plan** dialog box, enter a name and optional description for the plan.
4. Select **Hadoop** plan type.
5. Click **Next**.
6. To add the masking rule to the plan, click **Add Masking Components**.
7. Select the credit card masking rule to add to the plan. Click **OK**.
8. Click **Next**.
9. To skip adding the groups, click **Next**.
10. Review the masking component.
11. Click **Next**.
12. Configure the Oracle source connection and the Hive target connection.
13. Configure target properties, error and recovery settings, and advanced settings.

    The following image shows the Hadoop plan settings:



14. Click **Next**.
15. To override plan settings, click **Override Plan Settings** and enter the properties.
16. To override table settings, click **Override Data Source Settings** and enter the properties.
17. Click **Finish**.

Step 7. Generate and Run the Workflow

After you create the Hadoop plan, generate and run the workflow to populate the masked data in the Hive database.

1. In the project, click **Execute** to access the plans in the project.
2. Select the Hadoop plan that you created.
3. Click **Actions** > **Generate Workflow**.

   The **Generate Workflow** dialog box appears.
4. Select **Schedule Now**.
5. Click **Generate Workflow**.

   View the status of the workflow generation in the **Monitor** tab.
6. In the project, click **Execute** to access the plans in the project.
7. Click the Hadoop plan that you created.

   The plan opens in a separate tab.
8. Click **Actions** > **Execute Workflow**.

   The **Execute Workflow** dialog box appears.
9. Select the Data Integration Service.
10. Select **Schedule Now**.
11. Click **Execute Workflow**.

    View the status of the workflow run in the **Monitor** tab.

Author

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